

# Exploratory Data Analysis on the Forbes Richest Athletes Dataset

Report

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# 1.Introduction

This dataset gives the top ten athletes and their earnings from 1990 up to 2020. This dataset is a compilation of the Forbes list that appear every year on the highest earning athletes. Forbes also have other lists that are well known, like the list of richest people in the world. These lists are generally just meant for entertainment.

In this dataset, there are 301 entries. We expect to find 300 entries for the 30 years, however we have two athletes at number 10 in 2002. Some years have a few athletes sharing number 8 or 9, but then the total amount of athletes for that year is limited to ten. Although there are 301 entries, there are only 79 athletes that appear on this list.

The dataset has eight columns:

```
Index(['s.no', 'name', 'nationality', 'current rank', 'previous year rank',  
      'sport', 'year', 'earnings ($ million)'],  
      dtype='object')
```

These eight columns succinctly give the names, nationality, ranking and earnings in American dollar for each athlete for a particular year over the period of 30 years. It also includes the sport that they played and how/if they ranked the previous year. (The first column is the ID key for the data frame.)

This report has two parts. In the first part the cleaning of the dataset is discussed. The second part the clean data set is used for an exploratory data analysis.

## 2.Data Cleaning and Missing Data

Using pandas to look up the information on the data, we see that there are 301 entries for all the columns except for one – Previous Year Rank – that have 24 entries missing.

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 301 entries, 0 to 300
Data columns (total 8 columns):
#   Column                Non-Null Count  Dtype
---  -
0   S.NO                  301 non-null    int64
1   Name                  301 non-null    object
2   Nationality           301 non-null    object
3   Current Rank          301 non-null    int64
4   Previous Year Rank     277 non-null    object
5   Sport                 301 non-null    object
6   Year                  301 non-null    int64
7   earnings ($ million)  301 non-null    float64
dtypes: float64(1), int64(3), object(4)
memory usage: 18.9+ KB
```

Besides this one column, it is very promising to work with the dataset that is completed to this degree. It means there can be credible deductions. Furthermore, a check showed there were no duplicate rows.

In the first stages of cleaning, no columns were dropped because all of them could potentially be relevant. There were also no issues identified with the following four columns: S.no, Current Rank, Year, and Earnings. The cleaning of the other four columns are discussed below.

### 2.1. The 'Name' column

Looking at a sorted list of the unique names in the data frame, minor inconsistencies occur.

```
[ 'Aaron Rodgers', 'Aaron Rogers', 'Alain Prost', 'Alex Rodriguez', 'Andre Agassi', 'Andrew Luck', 'Arnold Palmer', 'Ayrton Senna', 'Buster Douglas', 'Cam Newton', 'Canelo Alvarez', 'Carson Wentz', 'Cecil Fielder', 'Conor McGregor', 'Cristiano Ronaldo', 'Dale Earnhardt', 'Dale Earnhardt Jr.', 'David Beckham', 'Deion Sanders', 'Dennis Rodman', 'Donovan "Razor" Ruddock', 'Drew Brees', 'Eli Manning', 'Emmit Smith', 'Evander Holyfield', 'Floyd Mayweather', 'Gary Sheffield', 'George Foreman', 'Gerhard Berger', 'Grant Hill', 'Greg Norman', 'Jack Nicklaus', 'Jacques Villeneuve', 'James Harden', 'Jeff Gordon', 'Jim Courier', 'Joe Montana', 'Joe Sakic', 'Jordan Spieth', 'Kevin Durant', 'Kevin Garnett', 'Kimi Raikkonen', 'Kirk Cousins', 'Kobe Bryant', 'Lance Armstrong', 'LeBron James', 'Lennox Lewis', 'Lewis Hamilton', 'Lionel Messi', 'Manny Pacquiao', 'Matt Ryan', 'Matthew Stafford', 'Michael Jordan', 'Michael Moorer', 'Michael Schumacher', 'Michael Vick', 'Mike Tyson', 'Monica Seles', 'Muhammad Ali', 'Neymar', 'Nigel Mansell', 'Novak Djokovic', 'Oscar De La Hoya', 'Patrick Ewing', 'Peyton Manning', 'Phil Mickelson', 'Rafael Nadal', 'Riddick Bowe', 'Roger Federer', 'Ronaldo', 'Rory McIlroy', 'Russell Wilson', 'Sergei Federov', 'Shaq O'Neal', 'Shaquille O'Neal', 'Stephen Curry', 'Sugar Ray Leonard', 'Terrell Suggs', 'Tiger Woods', 'Tom Brady', 'Valentino Rossi', 'Wayne Gretzky']
```

Spellings for Aaron Rodgers, Dale Earnhardt and Shaquille O'Neal were corrected. 'Ronaldo' was replaced by his full name - 'Ronaldo Gaúcho'. However, Neymar was kept as is, because he is known mononymously.

## 2.2. The 'Nationality' column

A search into the unique entries of the nationality's column reveals only a slight problem:

```
array(['USA', 'Brazil', 'France', 'Australia', 'Canada', 'UK', 'Austria', 'Germany', 'Russia', 'Italy', 'Finland', 'Switzerland', 'Philippines', 'Portugal', 'Dominican', 'Argentina', 'Filipino', 'Spain', 'Serbia', 'Northern Ireland', 'Ireland', 'Mexico'], dtype=object)
```

'Dominican' and 'Filipino' refer to the names people are called when they come from the Dominican Republic and the Philippines. They were changed to the country names like all the other entries. Northern Ireland is not changed to either Ireland nor the UK, it is kept as is.

Here are the 21 countries in the data frame:

```
array(['USA', 'Brazil', 'France', 'Australia', 'Canada', 'UK', 'Austria',  
      'Germany', 'Russia', 'Italy', 'Finland', 'Switzerland',  
      'Philippines', 'Portugal', 'Dominican Republic', 'Argentina',  
      'Spain', 'Serbia', 'Northern Ireland', 'Ireland', 'Mexico'],  
      dtype=object)
```

## 2.3. The 'Previous Year Rank' column

The column 'Previous Year Rank' needs cleaning up. A look at the unique values found across the rows in the column, shows 11 strange entries: 'nan', '>30', '>40', 'not ranked', 'none', '>10', '>20', '?', '>14', '>100' and '??'.

```
array([nan, '8', '1', '>30', '4', '5', '12', '6', '3', '9', '17', '13',  
      '10', '>40', '19', '40', '7', '11', '30', '22', '20', 'not ranked',  
      '38', '2', '15', '14', '26', 'none', '18', '>10', '>20', '?', '24',  
      '21', '>14', '>100', '??'], dtype=object)
```

One option to clean up this column by placing the ranking into bins – top 10, top 20, and so on. This gives us a general overview of where the athletes placed before they made the top 10 in that particular year. Here is a cleaned-up version for 'Previous Year Rank':

```
array(['not ranked', '>10', '>30', '>20', '>40', '>100'], dtype=object)
```

However, since there is also a lot of data entries missing from this column, it is not recommended to use in an analysis. The inconsistent entries together with the missing entries would make the data inviable for an accurate analysis. This column was therefore dropped.

## 2.4. The 'Sports' column

The column 'Sport' shows that most entries were not consistent.

```
array(['boxing', 'auto racing', 'golf', 'basketball', 'Basketball',  
      'Boxing', 'Auto Racing', 'Golf', 'Tennis', 'NFL', 'Auto racing',  
      'NBA', 'Baseball', 'Ice Hockey', 'American Football / Baseball',  
      'tennis', 'ice hockey', 'F1 Motorsports', 'NASCAR', 'Hockey',  
      'Auto Racing (Nascar)', 'F1 racing', 'American Football', 'soccer',  
      'baseball', 'cycling', 'motorcycle gp', 'Soccer', 'MMA'],  
      dtype=object)
```

There is inconsistent use of upper and lower case, and the use of abbreviations like 'NFL' or 'NBL' instead of the sport names. There are also multiple names used for auto racing.

There is one instance an athlete has two sports listed, 'American Football / Baseball.' However, a quick Google search shows that this athlete, Deion Sanders, played American football for the Colorado Buffaloes. Another unique case is Joe Sakic. Under sports only 'hockey' is listed. But after researching this athlete, it was changed to 'ice hockey.'

Here is an overview of the cleaned-up sports column:

```
array(['boxing', 'auto racing', 'golf', 'basketball', 'tennis',  
      'american football', 'baseball', 'ice hockey', 'soccer', 'cycling',  
      'motorcycle gp', 'mma'], dtype=object)
```

This concludes cleaning the dataset. The next section explores the data using visualisations.

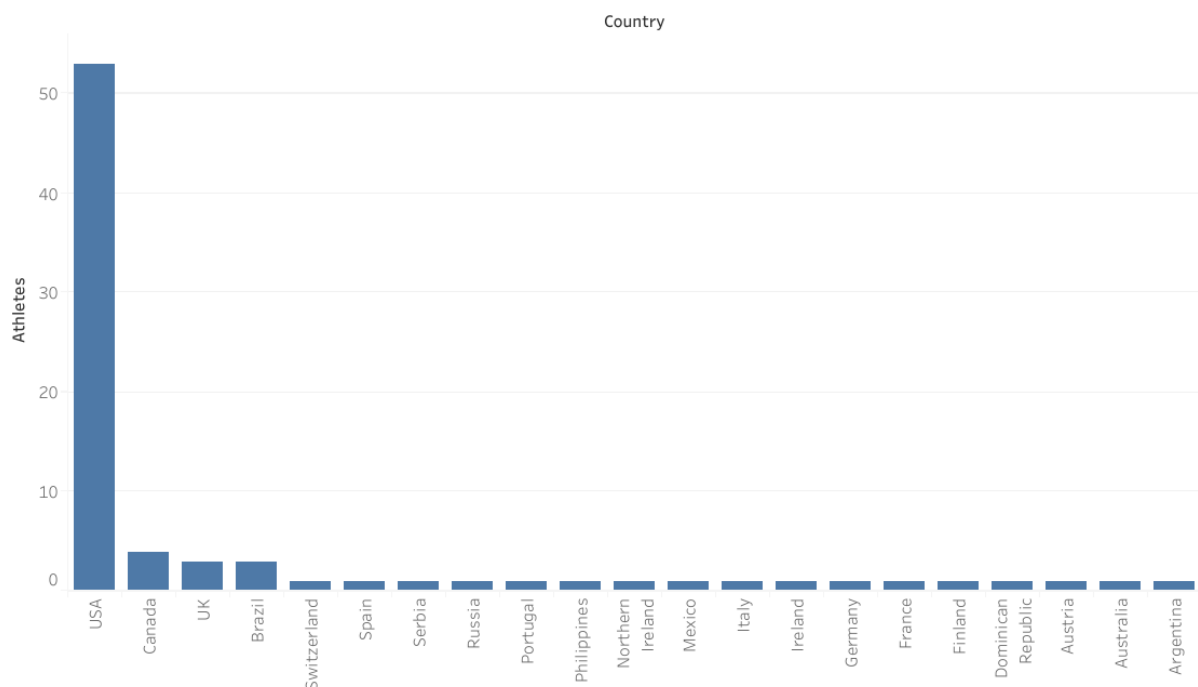
### 3. Exploratory Data Analysis

The clean dataset was explored through data visualisations created in Tableau.

#### 3.1. A look at the countries

There are 21 countries represented in the data. Below we see that 53 of the 79 athletes come from the United States of America. Most of the other countries have only one athlete. Does this mean that most of the money made in sport is made in America?

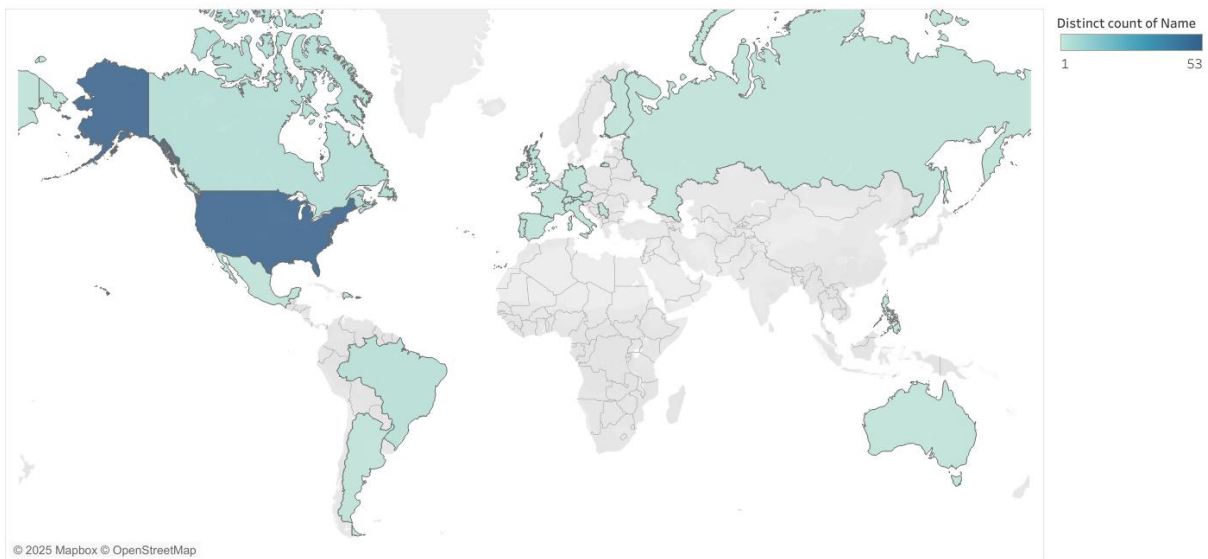
Nationalities of athletes



The publisher of the list, Forbes, is fundamentally an American company. Even though it is a highly reputable source, is it possible that the list is biased toward American athletes?

Here is a look at the countries represented on a map.

Countries of athletes



America is dark blue, because the most athletes hail from the USA. Other countries that have one or two athletes are in light blue. Nations with no athletes are grey. It is quite striking that most of the third world countries are coloured in grey.

Does that mean athletes from third world countries do not earn nearly as much as athletes from America or other first world countries? Or is the list bias towards the type of sports that interest Americans?

In the Forbes list there are no marathon runners, cricket players, or rugby players. Does this indicate a gap in the data? Perhaps further data collection should include the top 50 athletes and not just the top 10. Maybe data on other high-earning players from other sports should also be sourced in order to compare it to the Forbes list.

## 3.2. A look at the athletes

Here is a table showing the top ten athletes that appeared on the list the most. It also includes their sport. Tiger Woods and Micheal Jordan both appear almost 20 times.

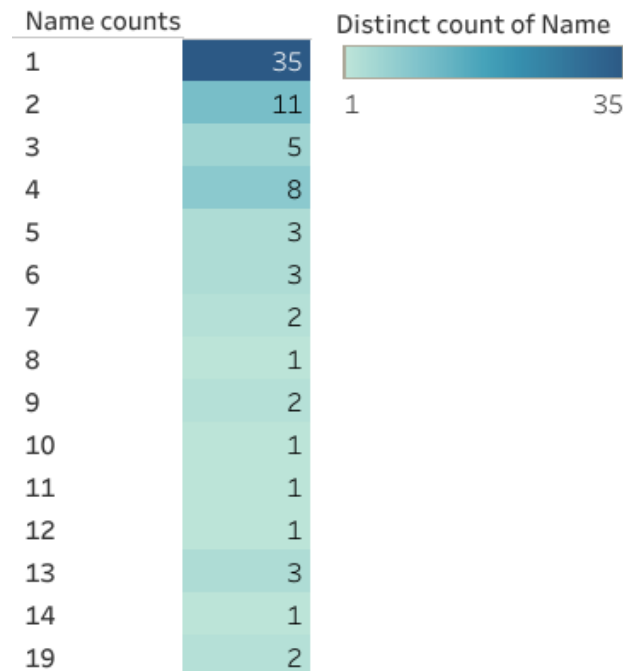
Most years on list

| Name               | Sport       |    |
|--------------------|-------------|----|
| Tiger Woods        | golf        | 19 |
| Michael Jordan     | basketball  | 19 |
| Kobe Bryant        | basketball  | 14 |
| Shaquille O'Neal   | basketball  | 13 |
| Michael Schumacher | auto racing | 13 |
| LeBron James       | basketball  | 13 |
| Roger Federer      | tennis      | 12 |
| Phil Mickelson     | golf        | 11 |
| Cristiano Ronaldo  | soccer      | 10 |
| David Beckham      | soccer      | 9  |



However, half of the athletes on the list only appear once in their career.

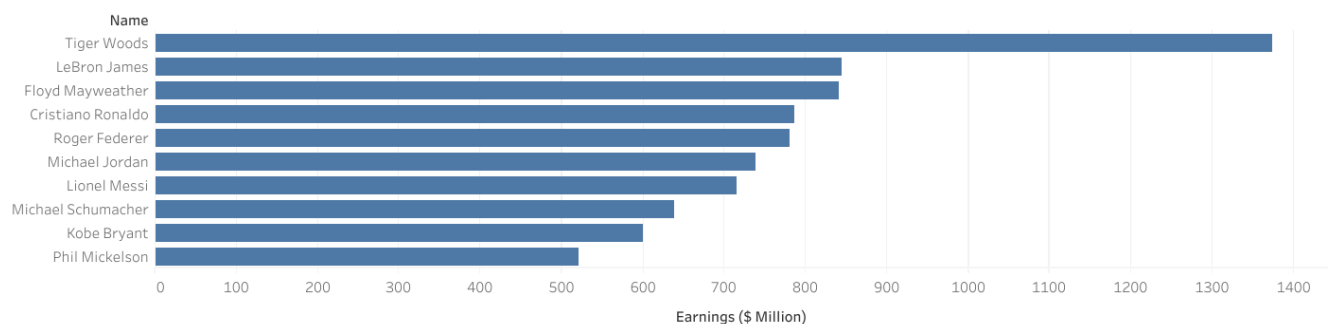
## Appearances on list



What does this mean? A guess is it shows how special an athlete must be to make it big.

Furthermore, we see that eight of the names that appear the most on the list, also earned the most amount of money in their careers. This is to be expected.

## Total of top earners

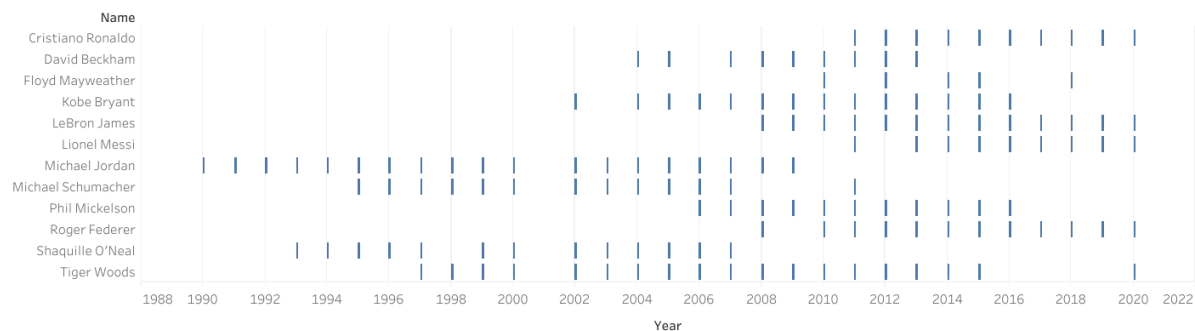


The two athletes that dropped out is Shaquille O'Neal and David Beckham. They were surpassed by Floyd Mayweather (5 appearances on list) and Lionel Messi (9 appearances on list). Besides Tiger Woods at number one (and an outlier), it is also interesting to note that the order in which they appear in this list is not the same as their appearances on the previous list.

The explanation for all of the above can be because of multiple factors. One of which is *when* they played. More money is earned in recent years than earlier years due to inflation.

Let us have a look at when these top earners played.

Years top earners played

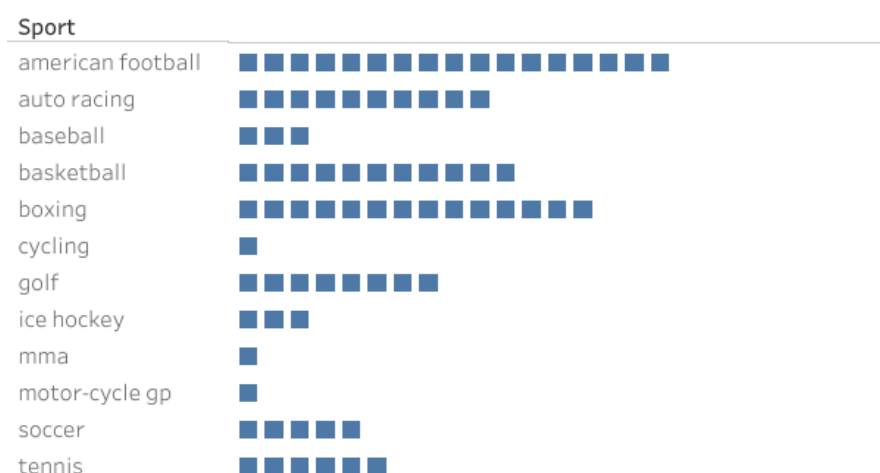


You can see that Messi and Mayweather are athletes that are recently active, whereas Shaquille O'Neal ended his career in 2007. It is also important to note that most of the top earners are active in the late 2000's and 2010's. The fact that Micheal Jordan, Shaquille O'Neal, and Micheal Shumacher still appear on this list, being of an older generation, just goes to show how highly they were paid. Note also that Tiger Woods has had a very long career and can continue to play golf at an older age – something that is not afforded to athletes from other sports. Tiger Wood is likely to keep on earning top dollar.

### 3.3. A look at the sports

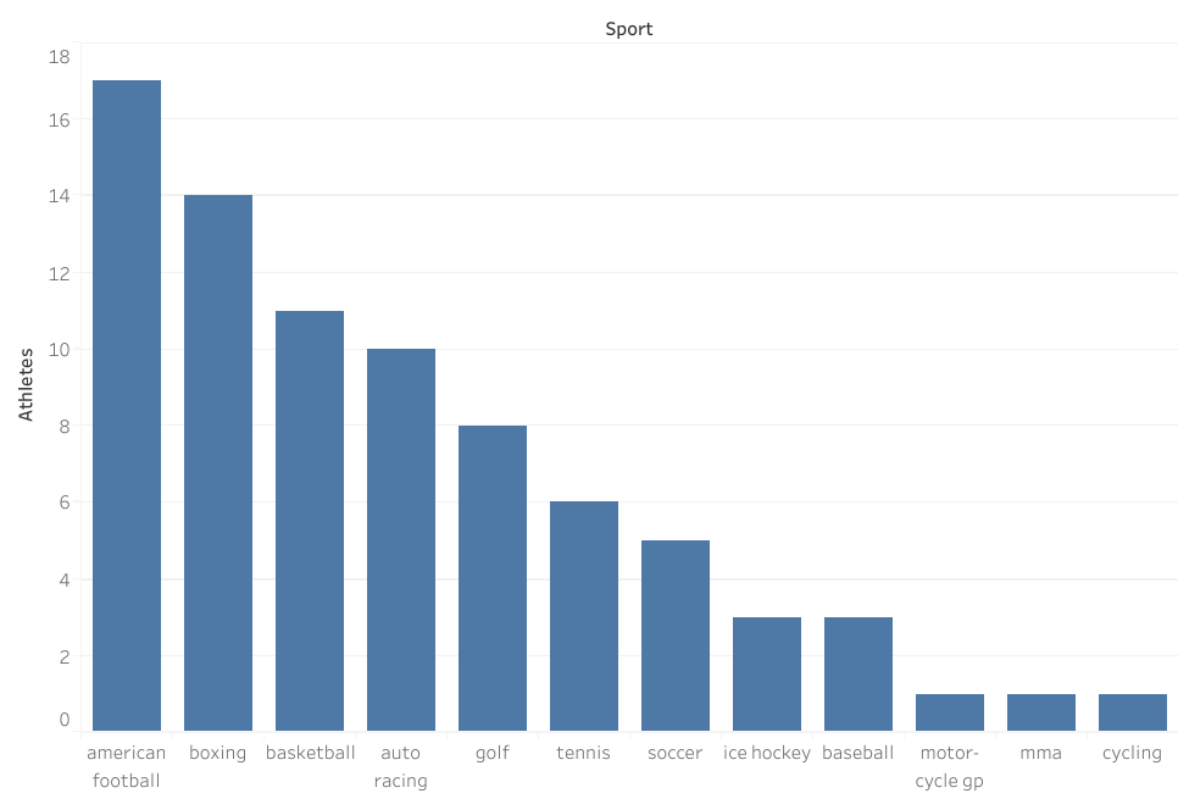
Twelve sports appear on the Forbes list. Most of the athletes on the list are American football players. However, this sport is not as popular anywhere else in the world as it is in America. Does this again indicate a bias in the data? Or does it mean that Americans spend a lot of money on football?

Athletes per sport



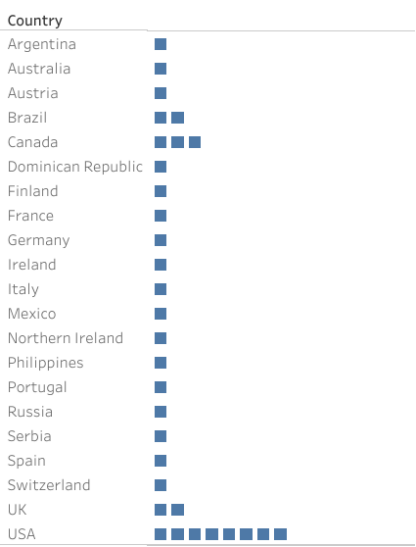
Here is the same data presented in another way to see the number of athletes per sport in a descending order.

Sports on list



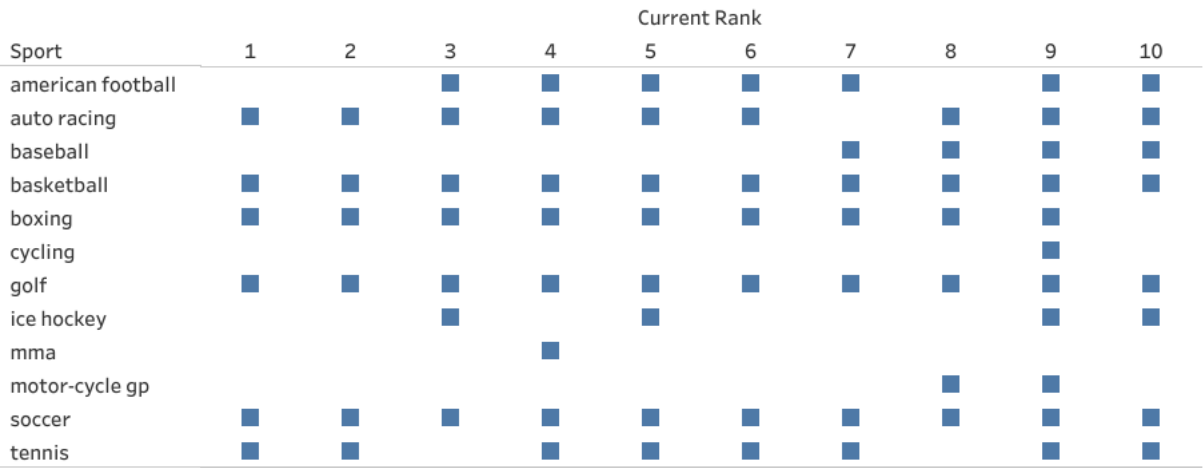
If we look at the sports from different countries, most countries supply athletes for only one sport. Most of the sports in this dataset are therefore popular sports in America. Does this indicate a bias in the data? Or that the sports that are popular in America are the highest paying?

Sports per country



Another way to look the sports is to see where they ranked. We see here that American football never ranked number one. These ballplayers were out paid by basketball, soccer, and boxing amongst others.

How sports usually rank

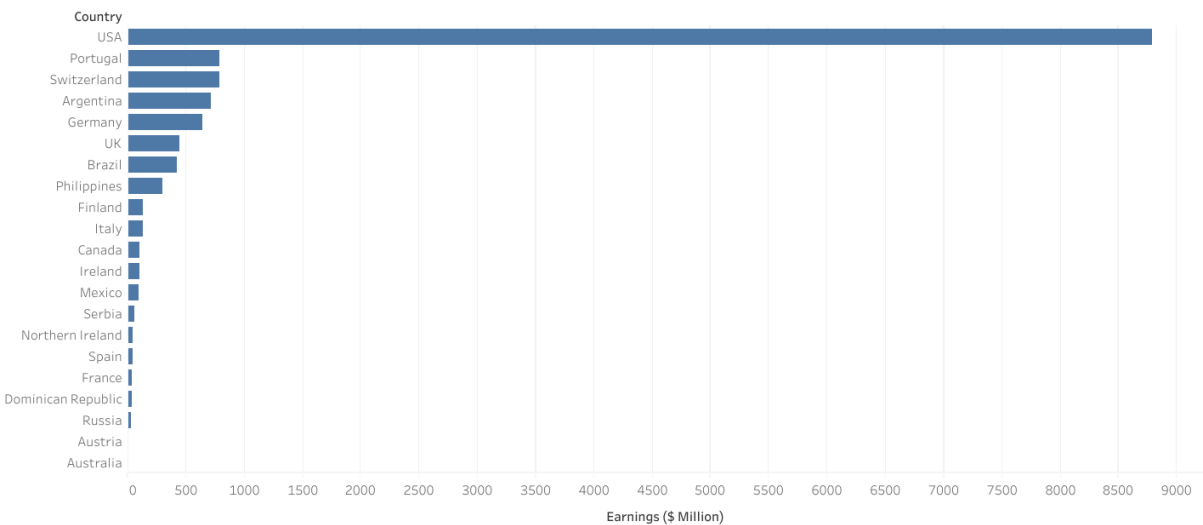


This can be explained because for ball sports like basketball and soccer athletes need a lot more refined ball skills that are very hard to come by. So, the supply and demand looks different to that of American football players who can get away with little less ball skills (but more muscle weight and strength).

3.4. A look at the earnings

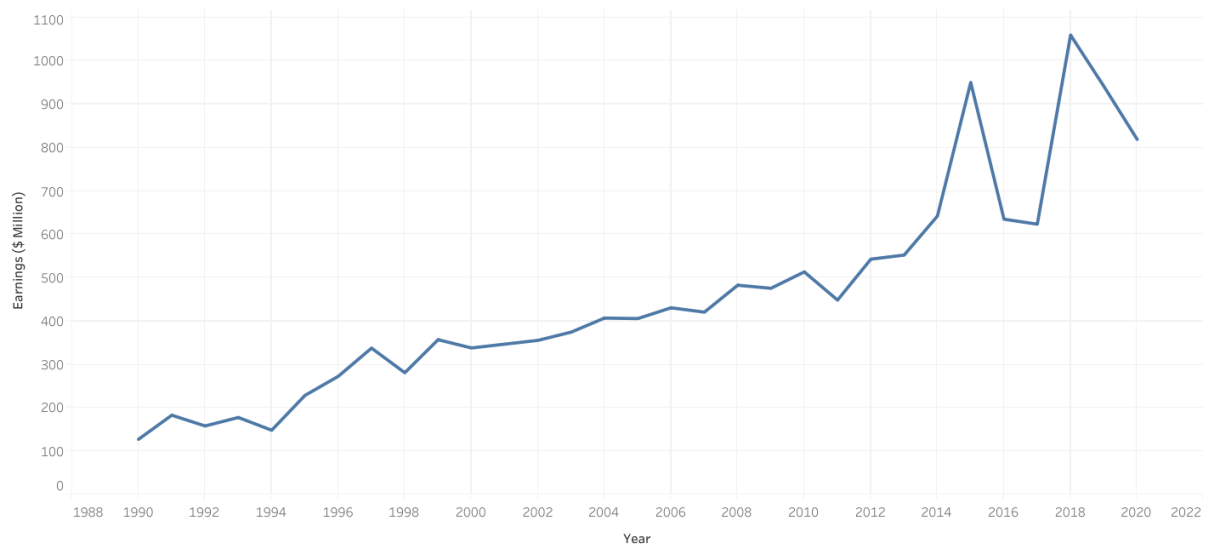
A visual shows that most of the money made from sport is in America. So much so that all the other countries are dwarfed. Austria and Australia are literally wiped out by the USA. It seems that if an athlete wants to make money, they should move to America.

Earnings per country



The next visual is a line graph to see how steady the earnings each year has risen. This is to be expected because of inflation.

Growth of earnings over 30 years



However, we see enormous spikes in 2015 and 2018. Both spikes are attributed to Floyd Mayweather. He earned a staggering 300 million in 2015. That is almost more than double what his closest peer earned – although his peer, Manny Pacquiao, also a boxer, earned a neat 160 million. In 2018 Mayweather earned 285 million, next up Lionel Messi with 111 million and Cristiano Ronaldo with 108 million. The late 2010s were a great time to be a boxer or soccer player.

## 4. Conclusion

This exploratory data analysis examined earnings among top-paid athletes as reported by *Forbes*. Key patterns emerge from the analysis: sports such as basketball and football feature prominently among the highest earners, reflecting their strong sponsorship ecosystems and media exposure. In addition, high-risk sports such as boxing and motor racing appear frequently among top earners, suggesting that sports with greater physical risk may also offer higher financial rewards, likely driven by pay-per-view models and elite-level scarcity.

It is important to acknowledge the limitations of this dataset. The heavy representation of American athletes and sports limits the generalisability of the results to global athletics. Furthermore, earnings figures are based on estimated values and do not fully capture differences in career length, financial stability, or long-term sustainability across sports.

Despite these limitations, the analysis is useful for understanding how income is distributed among elite athletes. A comparison between short, high-intensity careers and longer, more sustained earning trajectories highlights meaningful differences in financial outcomes. For example, while athletes like Floyd Mayweather achieved exceptionally high average annual earnings over a shorter career span, others such as Tiger Woods accumulated greater lifetime earnings over a much longer period, illustrating different pathways to financial success in professional sports.

Future analysis could benefit from incorporating non-US datasets, longitudinal earnings data, and additional variables such as career length, injury rates, and social media presence to provide a more comprehensive and globally representative view of athlete earnings.